[INCH-POUND]
A-A-59215
May 18, 1998
SUPERSEDING
J-L-231C
January 20, 1987

COMMERCIAL ITEM DESCRIPTION

LEADS, STORAGE-BATTERY

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE. This commercial item description (CID) covers low tension cables used for connections between storage batteries and other electrical equipment. The insulated leads are used for connections between the battery and equipment where heavy amperage 6-, 12-, and 24-volt (V) ungrounded circuits interconnect. The uninsulated leads and straps are used for bonding and grounding connections.

2. CLASSIFICATION. The leads shall be of the following types, classes, and styles, as specified (see 7.2):

Type I - Ground lead, uninsulating strap, flat woven or braided.

Style 1 - Battery to ground.
Style 2 - Ground to ground (bonding strap).

Type II - Battery lead, insulated (rope strand, concentric lay, or bunched).

Class A - Extra-flexible rope strand.
Class B - Stranded concentric lay or bunched.

Style 1 - Rope lay, concentric lay, or bunched with generator lead.
Style 2 - Rope lay, concentric lay, or bunched with lugs.
Style 3 - Rope lay, concentric lay, or bunched with straight universal post terminals.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 15E2), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6140

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

Style 4 - Rope lay, concentric lay, or bunched with angle universal post terminals.
Style 5 - Rope lay, concentric lay, or bunched with side terminal and lug.
Style 6 - Rope lay, concentric lay, or bunched with side terminals.

3. SALIENT CHARACTERISTICS.

3.1 Description. The storage battery leads shall consist of a length of low tension cable which is terminated at each end with a post terminal or terminal lug as required.

3.2 Standard commercial product. The storage battery leads shall, as a minimum, be in accordance with the requirements of this commercial item description and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this commercial item description but which are a part of the manufacturer's standard commercial product, shall be included in the storage battery leads being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Design. Insulator boots or caps shall be provided for type I, style 1 leads to provide an additional margin of safety during installation and servicing of the battery cable connections. Type II, class A or B leads, styles 1 through 6 shall be provided with insulator boots or caps to prevent inadvertent contact with energized conductors. Material used for construction of the cable or lead insulation shall be consistent with standard commercial practice.

3.3.1 Type I. Type I ground lead, uninsulated flat straps shall be made up of braided copper conductors and plated to resist corrosion.

3.3.1.1 Type I, style 1. Type I, style 1 ground lead shall be plated to resist corrosion and shall have a post terminal (see 3.3.5.1) on one end and a terminal lug (see 3.3.5.3) on the other end. The style 1 lead is used for connection from the battery to ground.

3.3.1.2 Type I, style 2. Type I, style 2 ground lead shall be uncoated and have a terminal lug (see 3.3.5.3) on each end for use from engine to ground.

3.3.2 Type II, insulated. Type II insulated lead shall be general purpose, insulated, braidless, and classified as extra flexible rope strand construction (class A), bunched or stranded concentric lay (class B).

3.3.2.1 Type II, class A. Type II, class A insulated extra flexible rope strand lead is used for connection between the battery and the starter, the battery and the solenoid, the starter and the solenoid, or between the battery and ground.

3.3.2.2 Type II, class B. Type II, class B insulated stranded concentric lay or bunched lead is normally used for connection between the battery and the starter, the battery and the solenoid, the starter and the solenoid, or between the battery and ground.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

3.3.3 Cable style. Type II, class A or B cable design shall be designated by the cable styles.

3.3.3.1 Type II, class A or B, style 1. Type II, class A or B, style 1 lead shall be insulated generator lead attached to a universal post terminal (see 3.3.5.1). The other end of lead shall be fitted with a terminal lug (see 3.3.5.3).

3.3.3.2 Type II, class A or B, style 2. Type II, class A or B, style 2 lead shall have a terminal lug (see 3.3.5.3) on each end. This style lead is used for connection between the solenoid and starter and similar applications.

3.3.3.3 Type II, class A or B, style 3. Type II, class A or B, style 3 lead shall have straight-neck post terminals (see 3.3.5.1) on each end. The post terminals used on these leads are the universal type fitting either a negative or positive battery post. This style lead is used as a connector when arranging batteries in series or parallel combinations.

3.3.3.4 Type II, class A or B, style 4. Type II, class A or B, style 4 lead shall have angle type post terminals (see 3.3.5.1). The post terminals used on these leads are the universal type arranged 180 degrees with respect to each other. This style lead is used as a connector when arranging batteries in series or parallel combinations.

3.3.3.5 Type II, class A or B, style 5. Type II, class A or B, style 5 lead shall have a side terminal with an integral bolt (see 3.3.5.2) on one end and a lug (see 3.3.5.3) on the other end. This style lead is used for connection between a side terminal battery and the starter or between the battery and ground.

3.3.3.6 Type II, class A or B, style 6. Type II, class A or B, style 6 lead shall have a side terminal with an integral bolt on each end (see 3.3.5.2). This style lead is used as a connector when arranging side terminal batteries in series or parallel combinations.

3.3.4 Insulation protection covering. When additional protective covering, such as a waterproof loom, is required for type II lead, the length of the loom and its location on the lead shall be as specified (see 7.2).

3.3.5 Termination. The termination for the leads shall conform to the requirements of 3.3.5.1 through 3.3.5.3, as applicable.

3.3.5.1 Lead terminals (battery post). The lead terminals for battery posts shall be the straight-neck type cast or molded from a corrosion-resistant metal. The terminal shall be of such design as to permit complete tightening with full contact on the battery post. A bolt and nut having a corrosion-resistant finish shall be provided with each terminal. The clamping bolt shall permit tightening to the degree necessary to provide a full power connection. The terminal shall be capable of providing a full power connection to either a positive or negative post.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

3.3.5.2 Lead terminals (side). Leads having side terminals shall be of correct design to connect to the electrical connection of a side terminal type battery, and shall have a corrosion-resistant finish. Each lead terminal shall include a bolt. The terminal shall be capable of providing a full power connection to either a positive or negative contact of a side terminal type battery.

3.3.5.3 Terminal lugs. Type II terminal lugs shall be formed from copper or brass alloy. The lug shall be plated or coated with lead-antimony alloy. Type I lead terminal lugs shall be compressed to fit closely around the flat ground strap. The connector shall be cadmium-plated or coated with lead-antimony alloy.

3.3.6 Mechanical strength. All completed cable leads shall be capable of withstanding a pull of 75 pounds (34 kilogram (kg)) for 1 minute.

3.4 Identification data. The lead assembly shall show inscribed on, or attached thereto, the manufacturer's name or trademark and part number. When specified (see 7.2), the lead assembly will also identify the manufacturer's production lot number.

4. REGULATORY REQUIREMENTS.

4.1 Materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this commercial item description are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this commercial item description.

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of ASTM SI-10 (IEEE/ASTM SI-10), and all other requirements of this commercial item description including form, fit and function are met. If a product is manufactured to metric dimensions and these dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformances.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

6. PACKAGING. The preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 The Federal Acquisition Regulation (FAR) is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 ASTM Standard is available from the Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

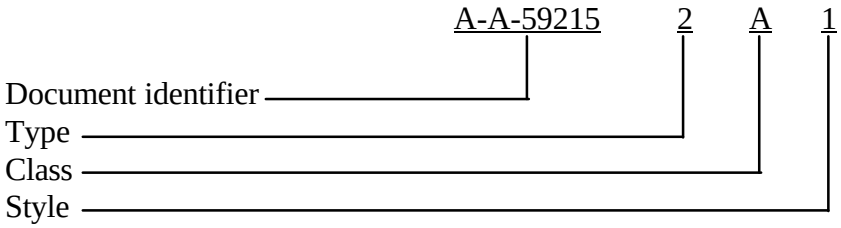
7.1.3 IEEE Standard is available from the Institute of Electrical and Electronics Engineers (IEEE), IEEE Service Center, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

7.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. Type, class, style, required (see 2).
c. When additional insulation protective cover is to be furnished, indicate the length and location on the lead (see 3.3.4).
d. When identification of the manufacturer's production lot number is required (see 3.4).

7.3 Supersession data. This CID replaces Federal Specification J-L-231C dated January 20, 1987.

7.4 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PINs to be used for items acquired to this description are created as follows:



The above PIN is for type II (battery lead), class A (extra-flexible rope strand), style 1 (rope lay). Note: Classes A and B are applicable to type II leads only.

7.5 Metric units. The values stated in either inch-pound units or SI units are to be regarded separately as standard. Within the text, the SI units are shown in brackets. The values stated in each system are not exact equivalents; therefore, each system should be used independently of the other. Combining values from the two systems may result in non-conformance with the standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

7.6 Subject term (key word) listing.

Battery
Cables
Ground
Storage batteries

7.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to previous issue due to the extensiveness of the change.

MILITARY INTERESTS:

Custodians:
Army - CR
Navy - YD1
Air Force - 99

Reviewer Activities:
Navy - AS, CG
Air Force - 03, 19, 33, 80, 82
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
Navy - YD1

(Project 6140-0874)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER**	**2. DOCUMENT DATE (YYMMDD)**
	A-A-59215	980518

3. DOCUMENT TITLE

LEADS, STORAGE BATTERY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *Include Area Code)* (1) Commercial	(2) AUTOVON
DWAYNE MILLS	805-982-9681	551-9681
c. ADDRESS *(Include Zip Code)* COMMANDING OFFICER, NCBC CODE 15E2M 1000 23RD AVENUE PORT HUENEME, CA 93043-4301	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil/Document Id: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.